The Quest for Depth: Achieving Convergence and Residual Learning

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Spring 2026

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https://dannagurari.colorado.edu/course/neural-networks-and-deep-learning-spring-2026/

Review

- Last lecture:
  - NN neglect: 2nd AI winter & retreat to shallow models with engineered features
  - NN hope lingers: the unfashionable quest for *depth*
  - NNs take over: AlexNet triggers a revolution where ‘deep learning’ dominates
  - Programming tutorial: AlexNet in PyTorch
- Assignments (Canvas)
  - Lab assignment 1 grades were released on Friday
  - Lab assignment 2 due in 1.5 weeks
    - For clarifications about common assignment questions, review the Canvas assignment page
- Questions?
  - Outside of questions for me, please post them to Piazza and/or attend office hours

Today’s Topics

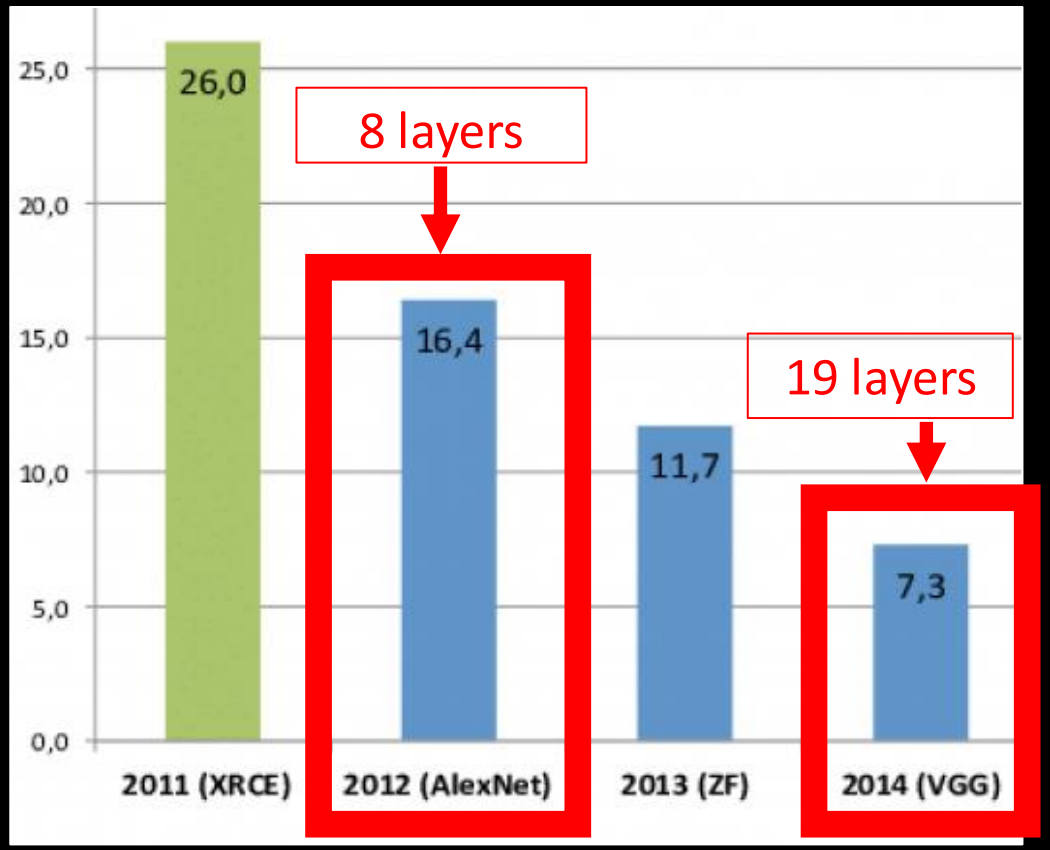
- The quest for deeper neural networks: why do they fail?
- Convergence is hard: He initialization, batch normalization, Adam optimization
- More is needed: residual learning with skip connections (ResNet)
- Programming tutorial

Today's Topics

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Context: In 2013 and 2014, Deeper CNNs Emerged as Secret Sauce for Achieving State-of-the-Art

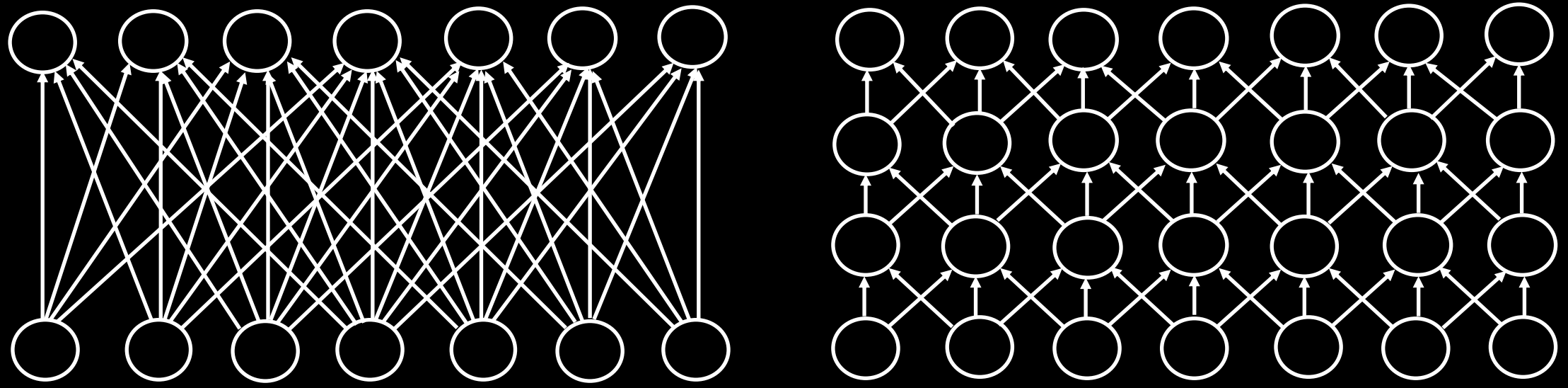
e.g., progress of models on ImageNet (Top 5 Error)



https://www.edge-ai-vision.com/2018/07/deep-learning-in-five-and-a-half-minutes/

e.g., VGG Idea: Smaller Convolutional Filters

- Replace larger filter with stack of smaller filters; e.g., replace 7x7 with three 3x3s

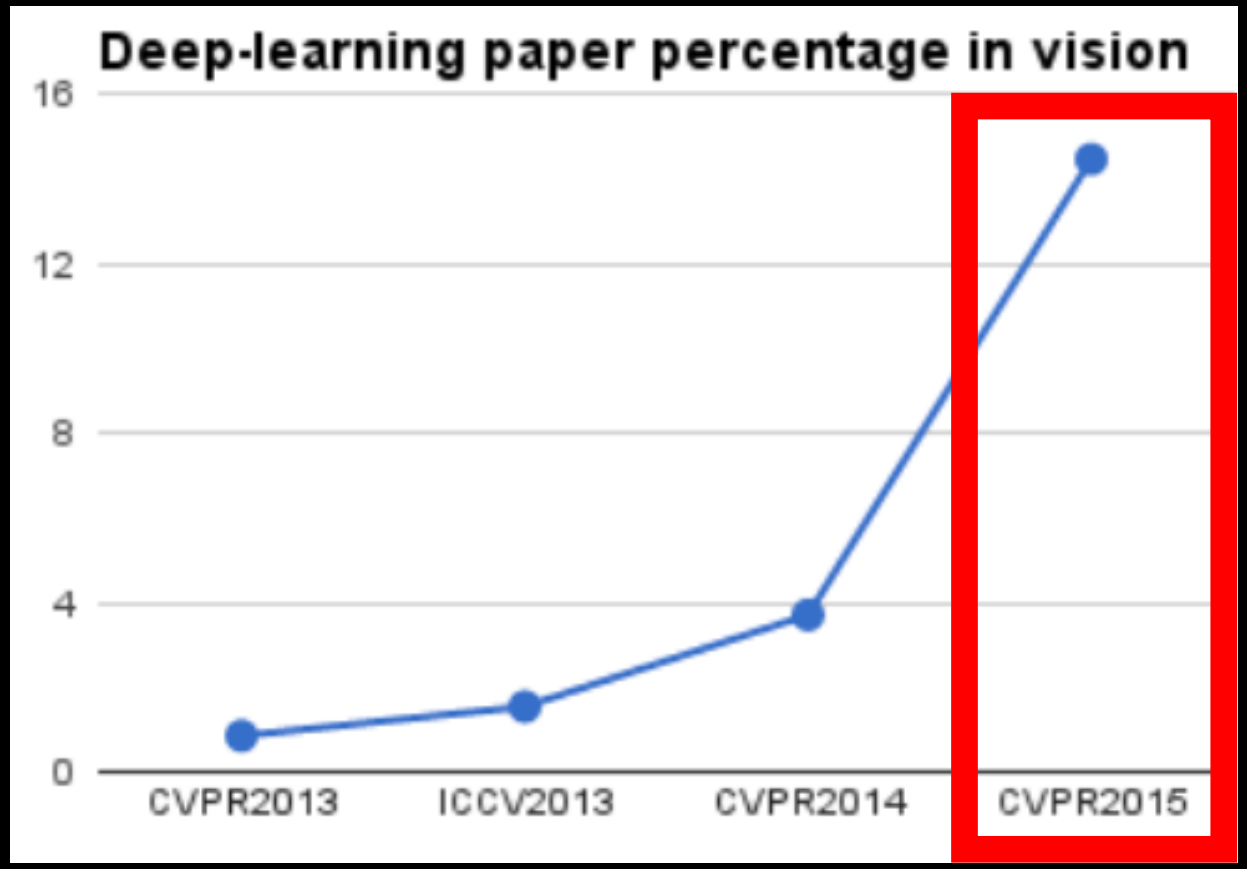


- Benefits:
  - Reduces # of parameters: multiple of 27 (3×3^2) parameters vs 49 (7×7) parameters
  - More discriminative classifier since more non-linear rectifications: 3 vs 1

Simonyan & Zisserman. Very Deep Convolutional Networks for Large-Scale Image Recognition. ICLR 2015.

Community Optimism and Excitement: What Tricks Support Going Deeper?

e.g., engagement in computer vision



https://www.slideshare.net/xavigiro/saliency-prediction-using-deep-learning-techniques

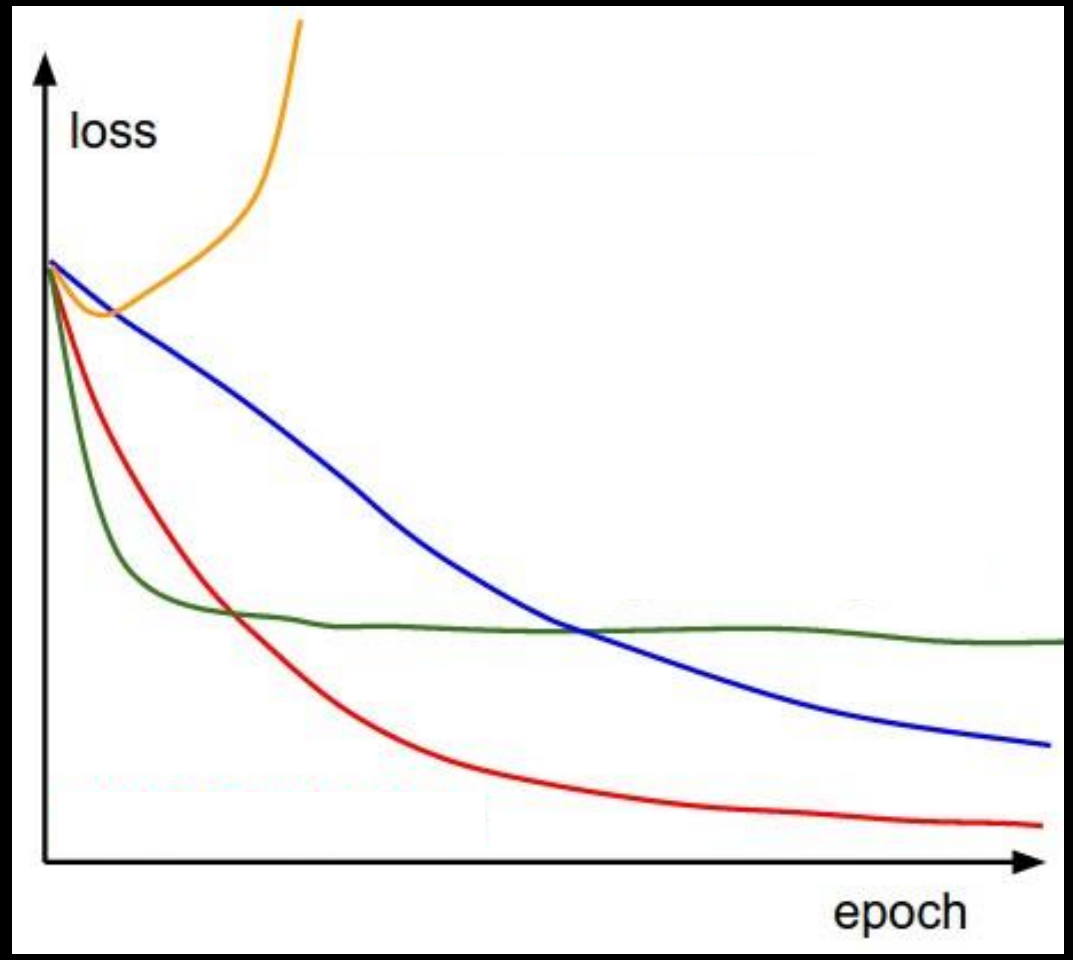
2015 was a pivotal year, with a wave of critical innovations that reshaped how we train deep networks!

Today's Topics

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Depth Amplified Issues for Effective Learning

Recall, we want a good loss curve during training:



https://cs231n.github.io/neural-networks-3/#update

(2015, ICCV)

Delving Deep into Rectifiers: Surpassing Human-Level Performance on ImageNet Classification

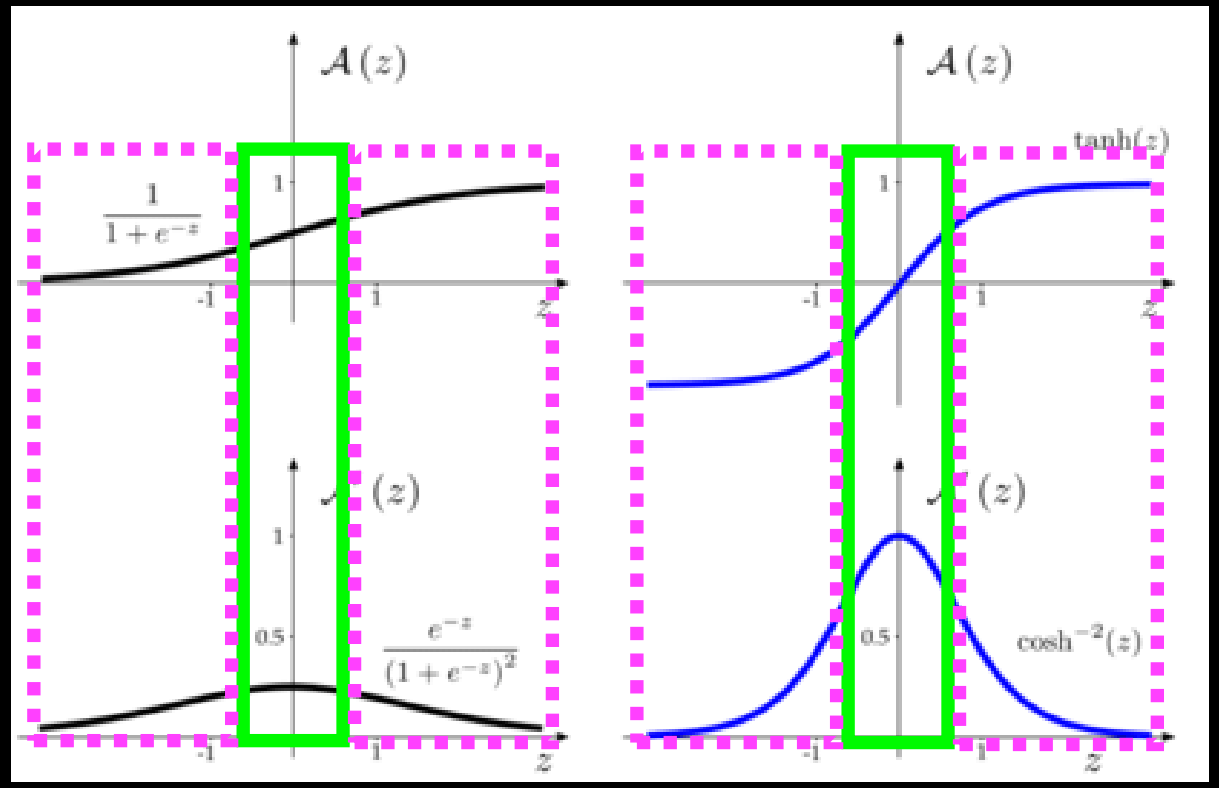
Kaiming He Xiangyu Zhang Shaoqing Ren Jian Sun

Microsoft Research

Recall: Initialize Weights to Establish Weighted Sums that are Good for Learning

Sigmoid

Tanh

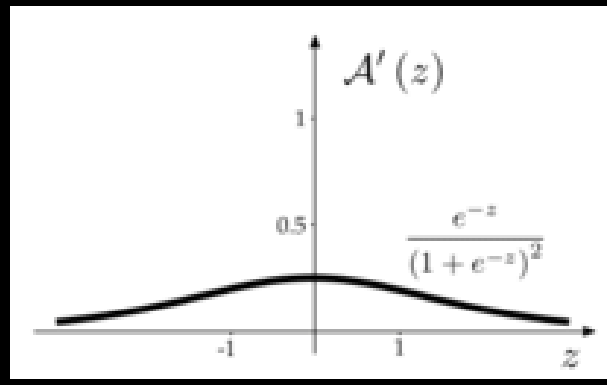


Masi et al. Journal of the Mechanics and Physics of Solids. 2021

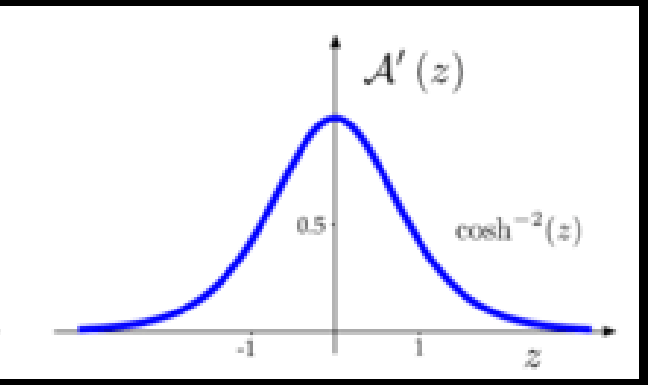
Recall: Initialize Weights to Establish Weighted Sums that are Good for Learning

Approach: randomly draw from distribution with mean 0 and standard deviation determined by number of incoming connections to a neuron ("fan-in"):

sigmoid

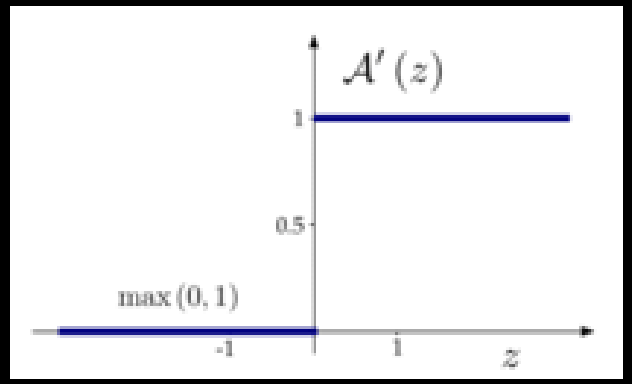


tanh



(mean ~0 and roughly equal positive and negative slopes preserves variance of the output)

ReLU



(roughly equal 0 and 1 slopes, cuts variance of output by half)

1998 "Efficient backprop" paper: (symmetric activation functions) $\sqrt{\frac{1.0}{m}}$

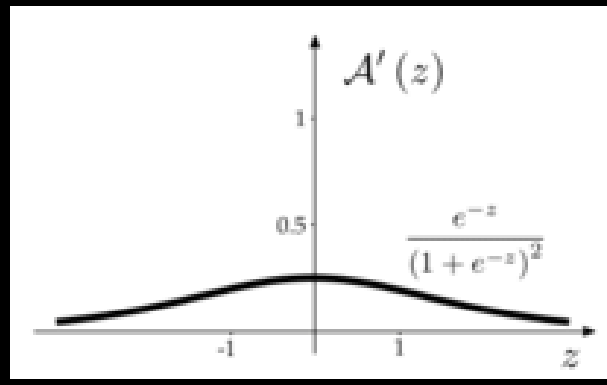
Problem: each layer reduces sum magnitude; stacked across layers, these reductions compound, causing activations and *gradients* to vanish for "deep" networks (e.g., ~30 layers)

Figure source: Masi et al. Journal of the Mechanics and Physics of Solids. 2021

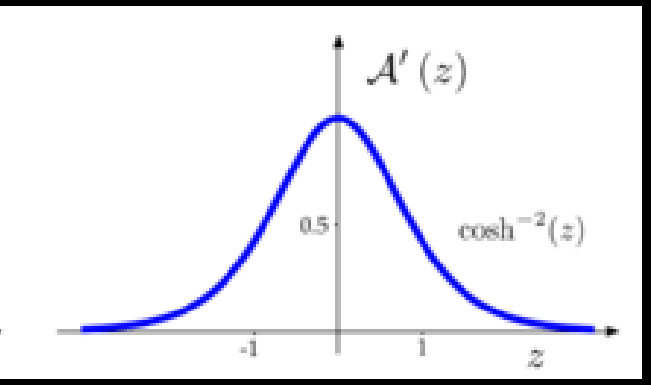
Idea: He Initialization for ReLU

Approach: randomly draw from distribution with mean 0 and standard deviation determined by number of incoming connections to a neuron ("fan-in"):

sigmoid



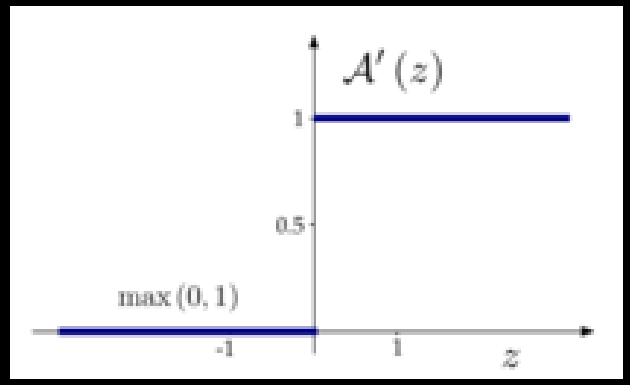
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1998 "Efficient backprop" paper: (symmetric activation functions) $\sqrt{\frac{1.0}{m}}$

ReLU



(roughly equal 0 and 1 slopes, cuts variance of output by half)

Initialization: $\sqrt{\frac{2.0}{m}}$

Why a value of 2?
- Preserves variance

Figure source: Masi et al. Journal of the Mechanics and Physics of Solids. 2021

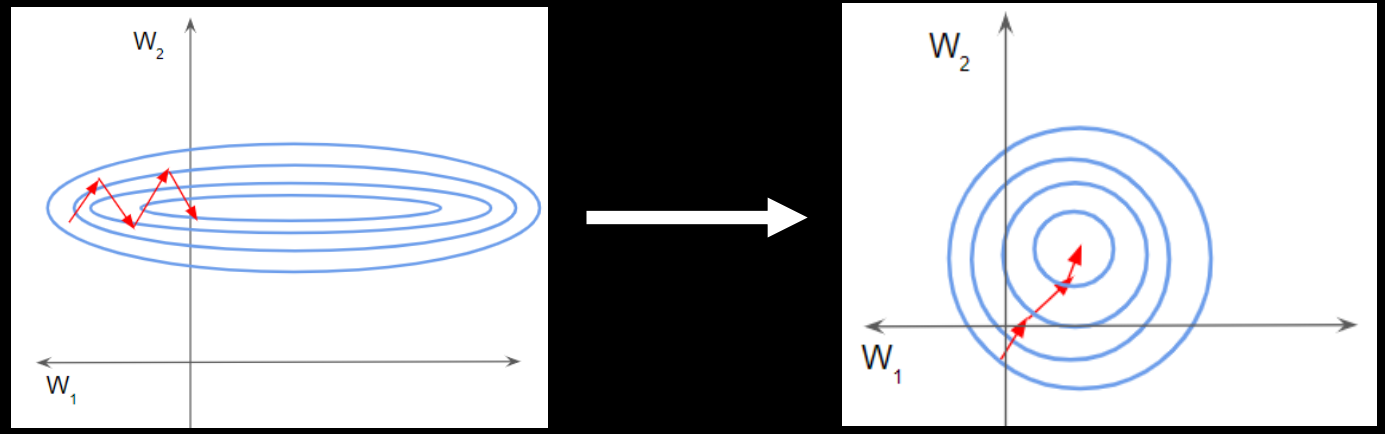
(ICML 2015)

Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift

Sergey Ioffe SIOFFE@GOOGLE.COM
Christian Szegedy SZEGEDY@GOOGLE.COM
Google, 1600 Amphitheatre Pkwy, Mountain View, CA 94043

Motivation: Recall (from Shallow Learning) that Normalizing Input Data Accelerates Learning

Approach: normalize input data to improve loss landscape



https://towardsdatascience.com/batch-norm-explained-visually-how-it-works-and-why-neural-networks-need-it-b18919692739

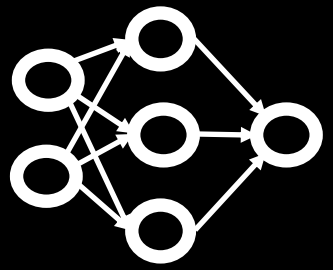
Intuition: enables driving in the optimal direction fast (rather than slower, indirect zig-zagging)



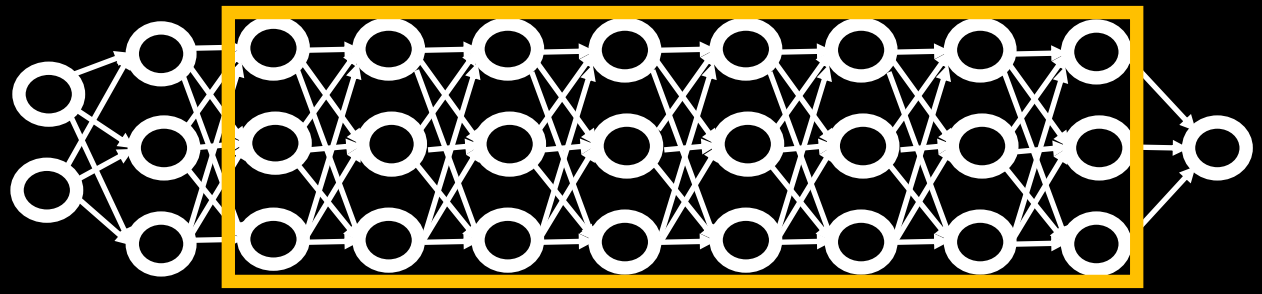
https://texomaexcavationandconstruction.com/gravel-roads-construction-forest-wooded-remote-areas/

Observation: What Changes for “Deeper” Networks?

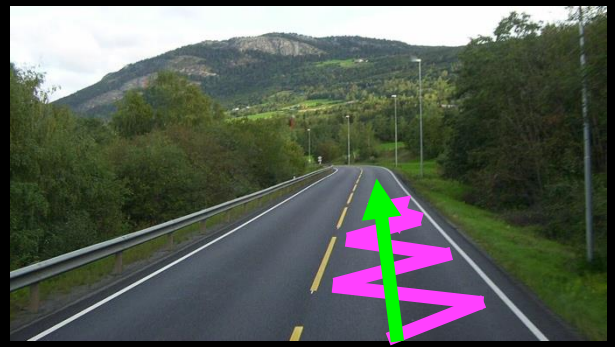
Shallow Network



Deeper Network



Problem: Input normalization targets only the first layer. Each layer’s weights rescale outputs, causing feature-scale differences to compound with depth. This, in turn, leads to inefficient zig-zagging for learning:

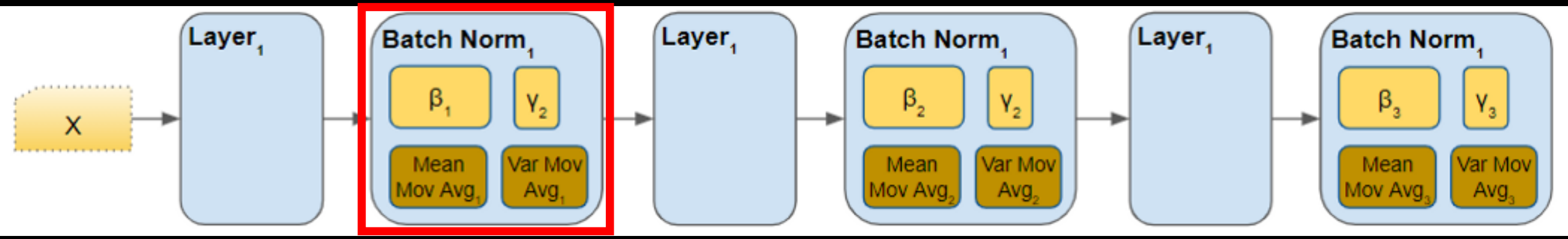


https://texomaexcavationandconstruction.com/gravel-roads-construction-forest-wooded-remote-areas/

Idea: add normalization for intermediate layers, so learning is fast for the entire network!

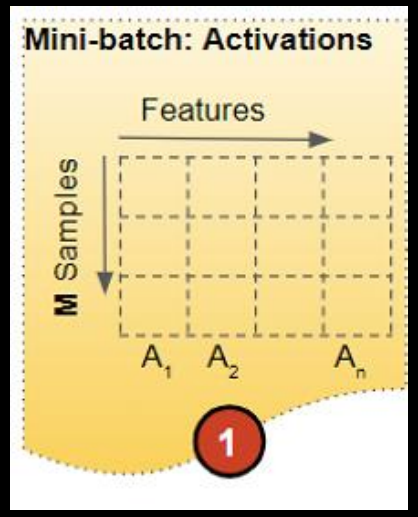
Approach: Add Layer with Learnable Parameters

A unique BN module is applied to activation values for intermediate layers!



https://towardsdatascience.com/batch-norm-explained-visually-how-it-works-and-why-neural-networks-need-it-b18919692739

Approach: Batch Normalization Layer

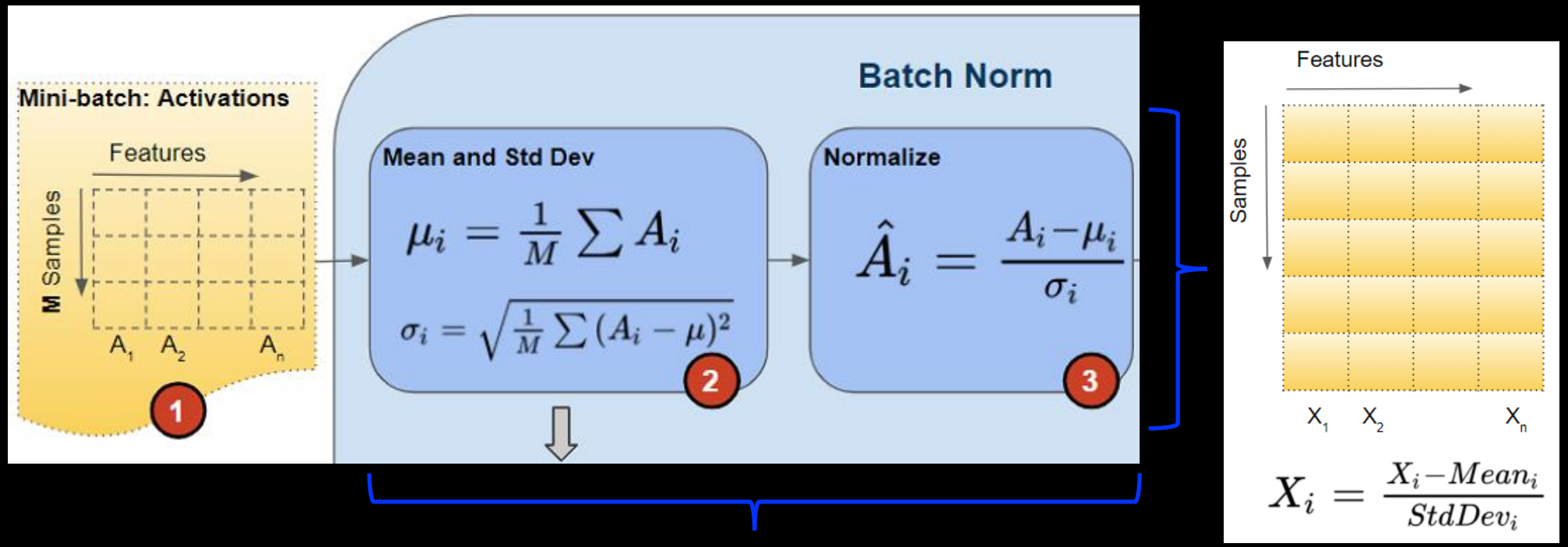


Input: mini-batch

- e.g., assume a fully connected layer, each row represents a unique example and each column represents all outputs (features) from one of the layer's neurons
- How many examples are in the toy example's mini-batch?
- How many neurons are leaving the toy example's layer?

https://towardsdatascience.com/batch-norm-explained-visually-how-it-works-and-why-neural-networks-need-it-b18919692739

Approach: Batch Normalization Layer (Training)

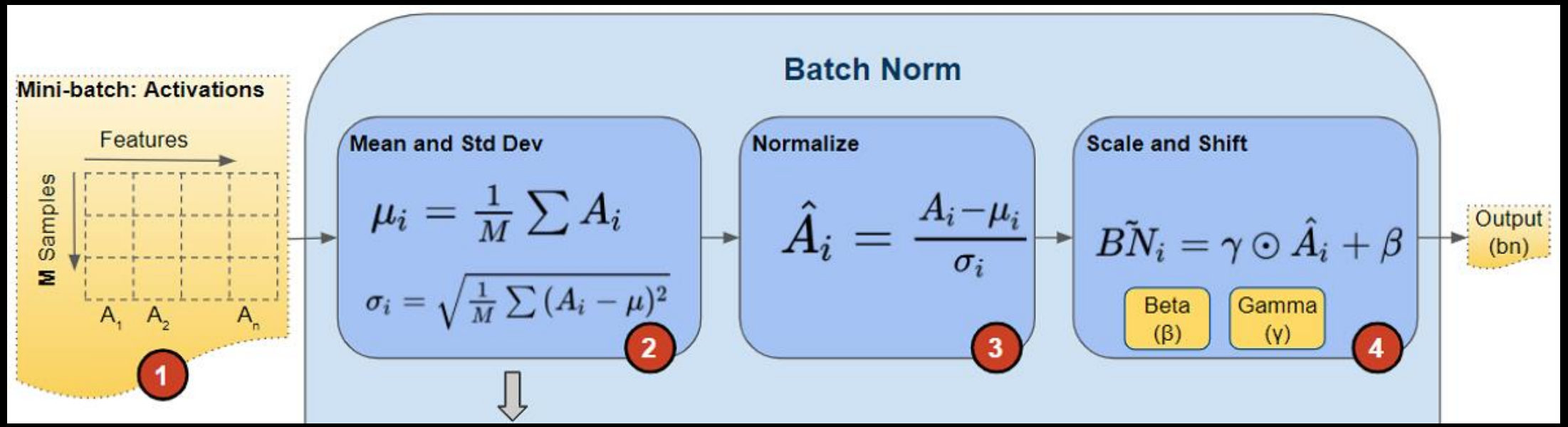


For every feature, compute mean and standard deviation values with all mini-batch examples and then use those values to modify the features to target distributions

Recall: using a mini-batch is an approximation of using the full training dataset for input normalization

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Approach: Batch Normalization Layer (Training)

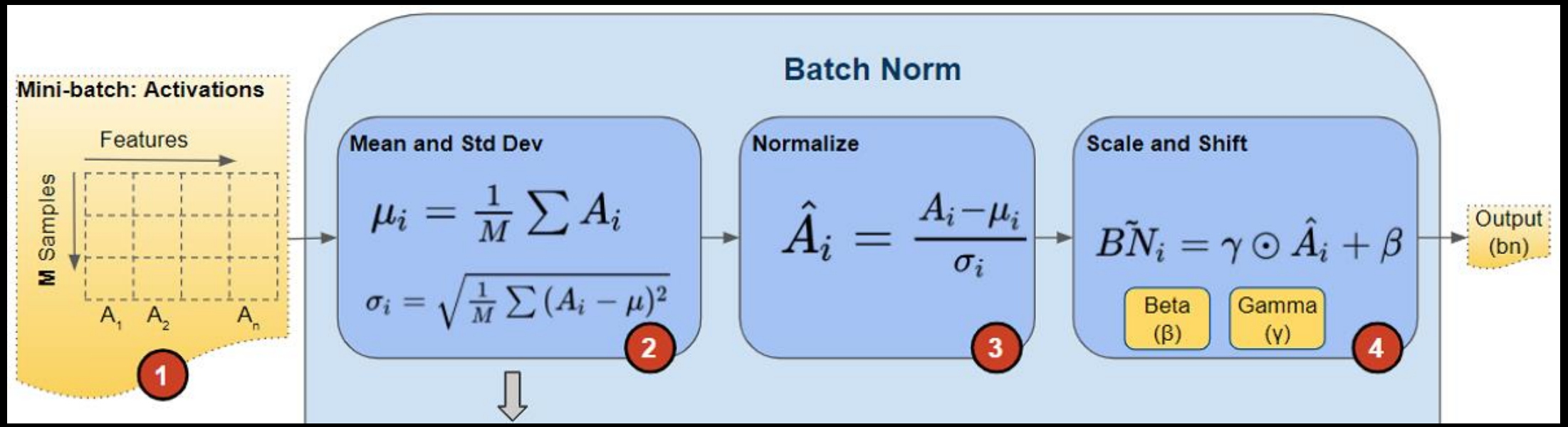


Two learned parameters preserve representational power by allowing the layer to recover its original activations;
- what values would recover the original parameters?

Beta obsoletes need for bias in Wx + b (subsumes that value)

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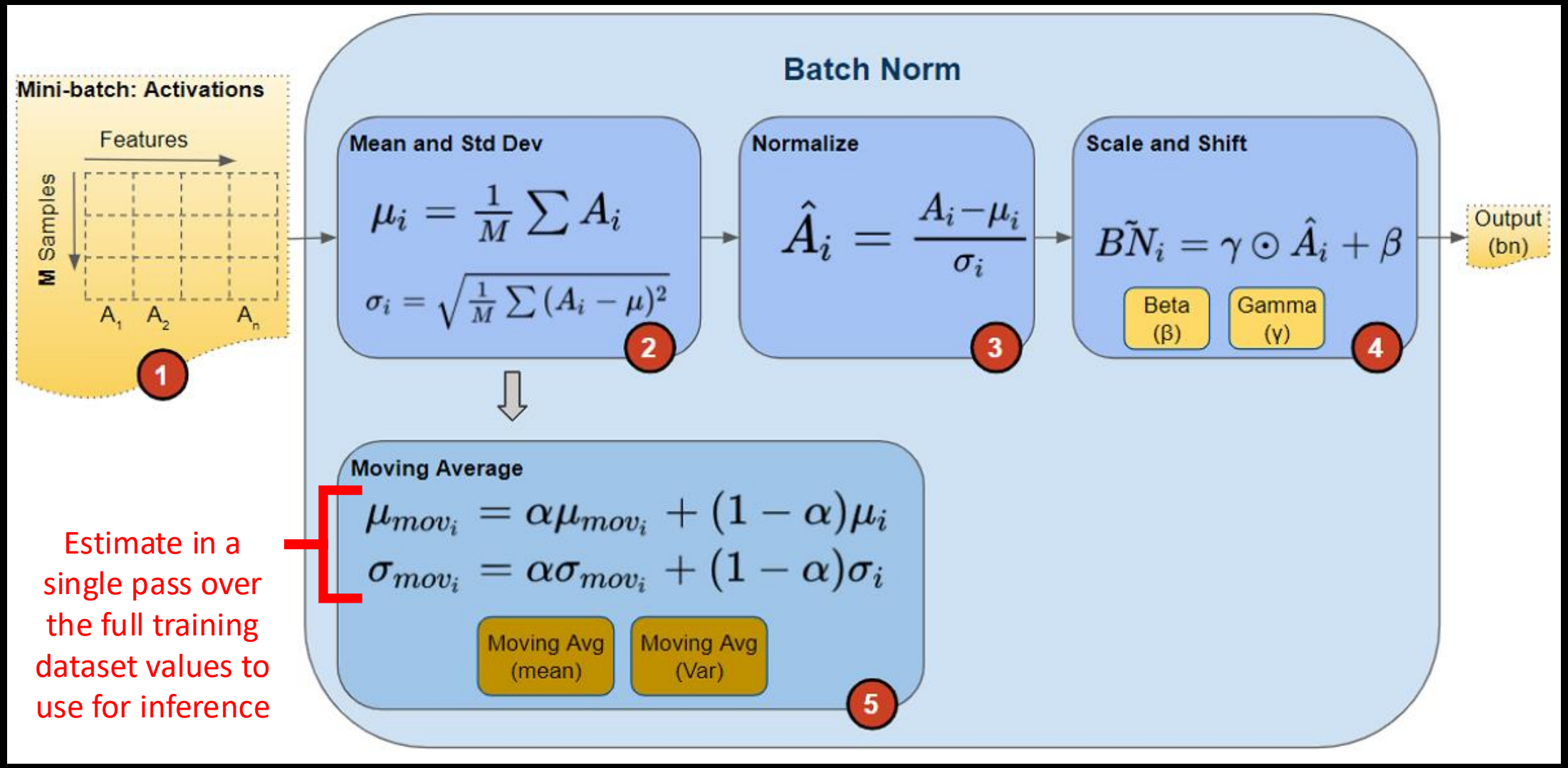
Approach: Batch Normalization Layer (Training)



For training, we can use the standard gradient descent approach because BN is a differentiable transformation!

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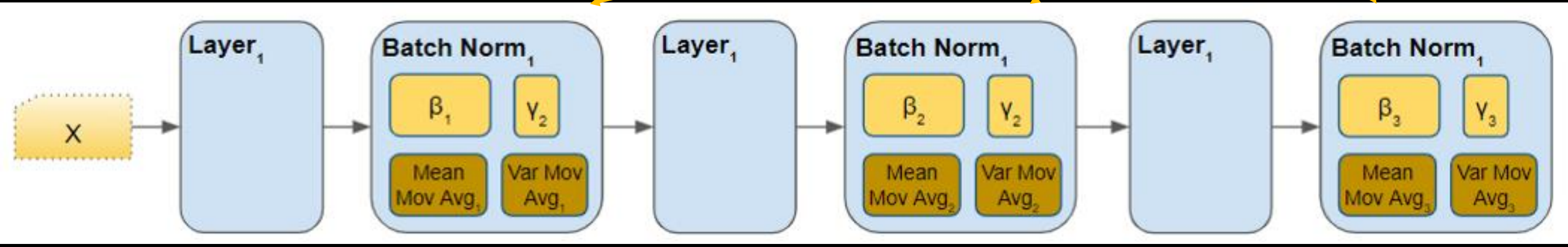
Approach: Batch Normalization Layer (Training)



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Approach: Add Layer with Learnable Parameters

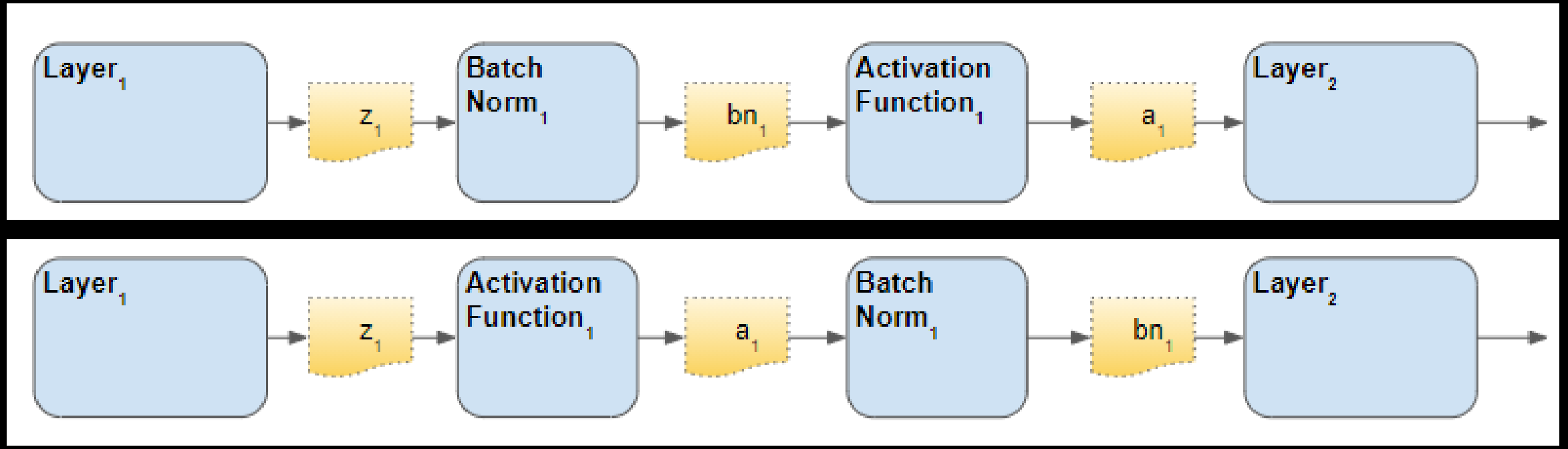
A unique BN module is applied to activation values for intermediate layers!



How many model parameters are trained for this subnetwork?

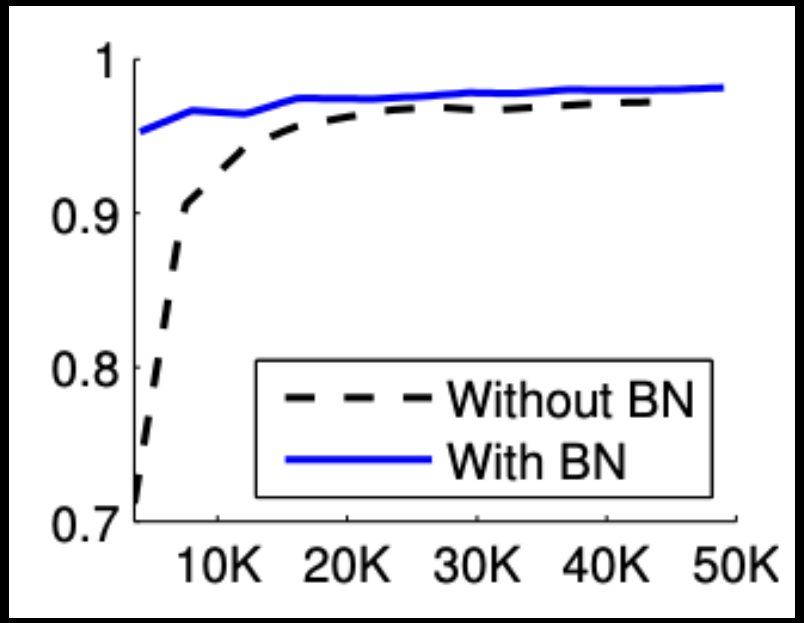
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Implementation Detail: BN Can Be Used Before and After Computing Activations



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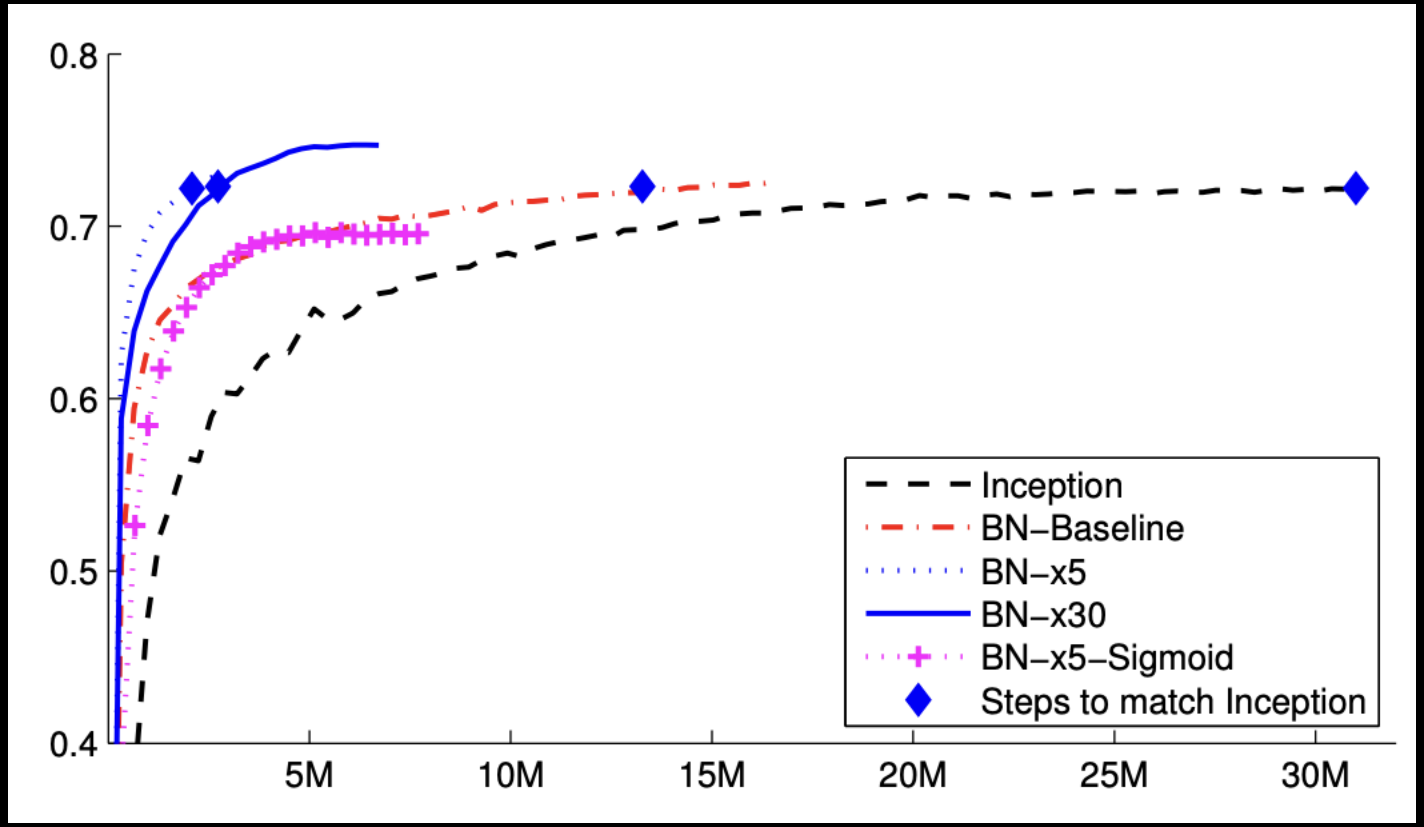
Experiments: BN Accelerates Learning!



Model with 3 fully-connected hidden layers and 100 neurons per layer with sigmoid activation functions benchmarked on MNIST (mini-batch size of 60 examples)

[Ioffe and Szegedy. ICML 2015]

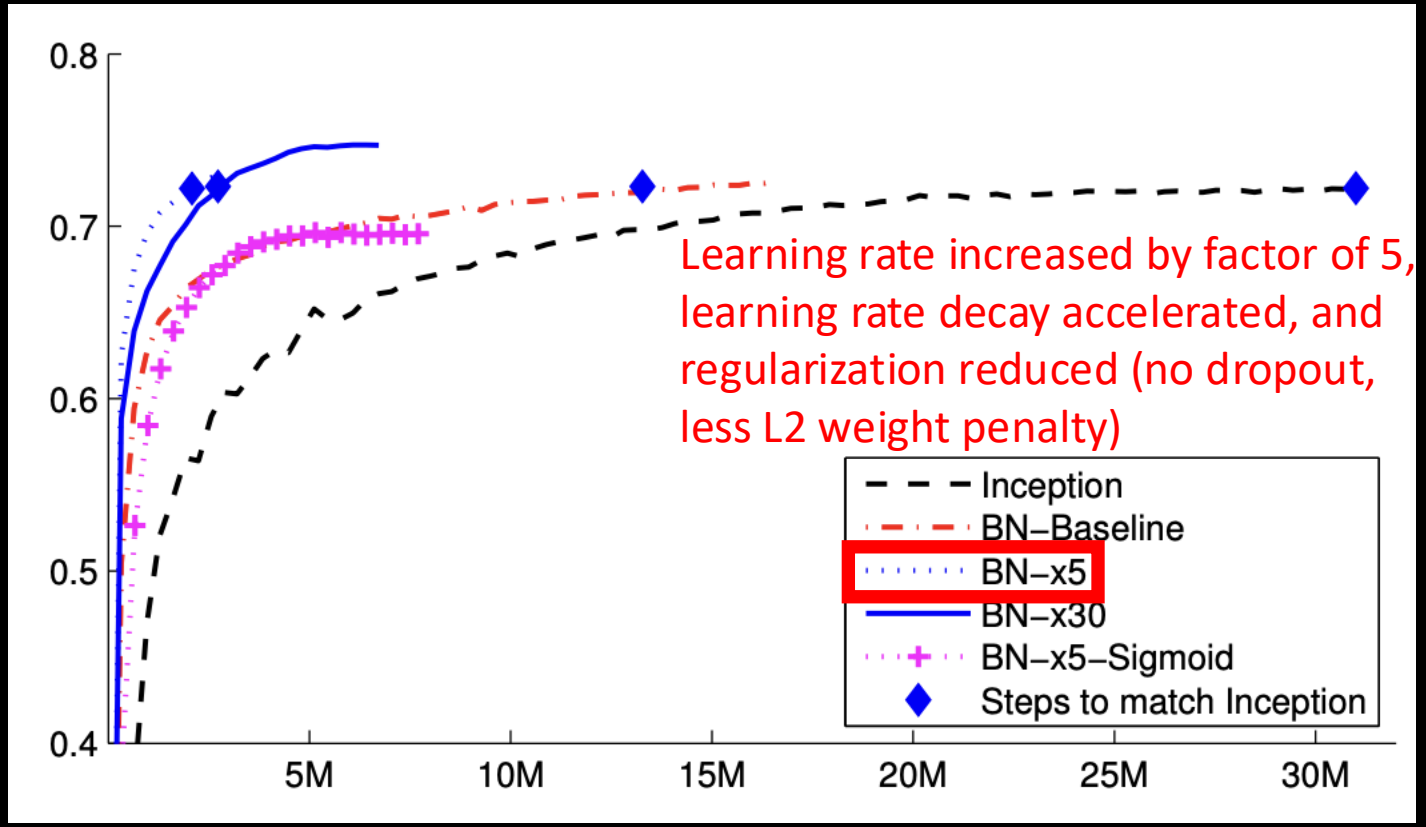
Experiments: BN Accelerates Learning!



Five BN variants extending deep SOTA ImageNet classifier at the time (SGD with momentum; mini-batch size of 32 examples)

[Ioffe and Szegedy. ICML 2015]

Experiments: BN Accelerates Learning!



Comparable performance achieved for ImageNet classification using only 7% of the training steps!

[Ioffe and Szegedy. ICML 2015]

Published as a conference paper at ICLR 2015

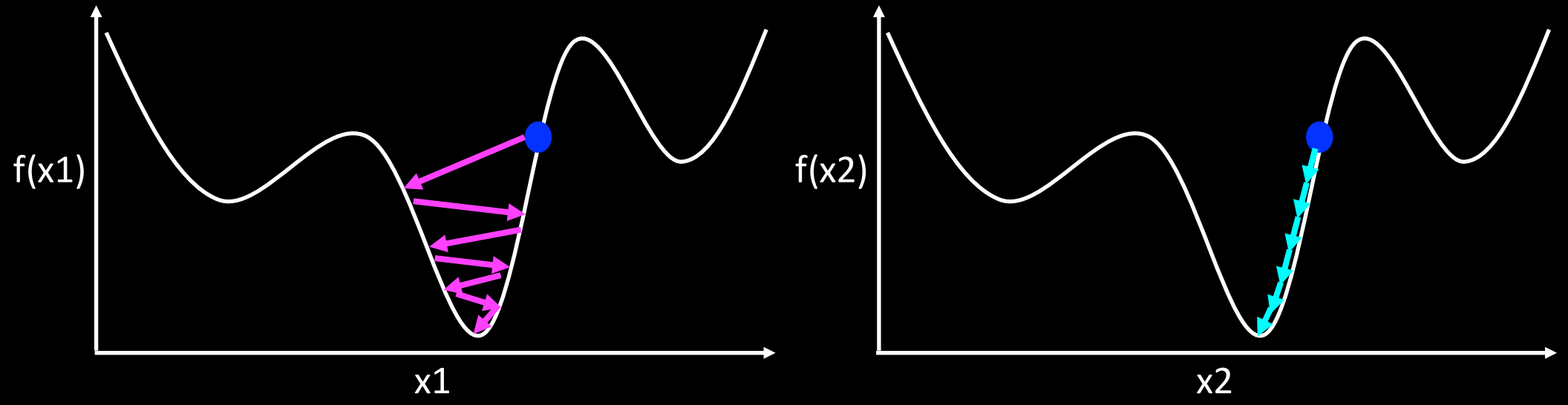
ADAM: A METHOD FOR STOCHASTIC OPTIMIZATION

Diederik P. Kingma*
University of Amsterdam, OpenAI
dpkingma@openai.com

Jimmy Lei Ba*
University of Toronto
jimmy@psi.utoronto.ca

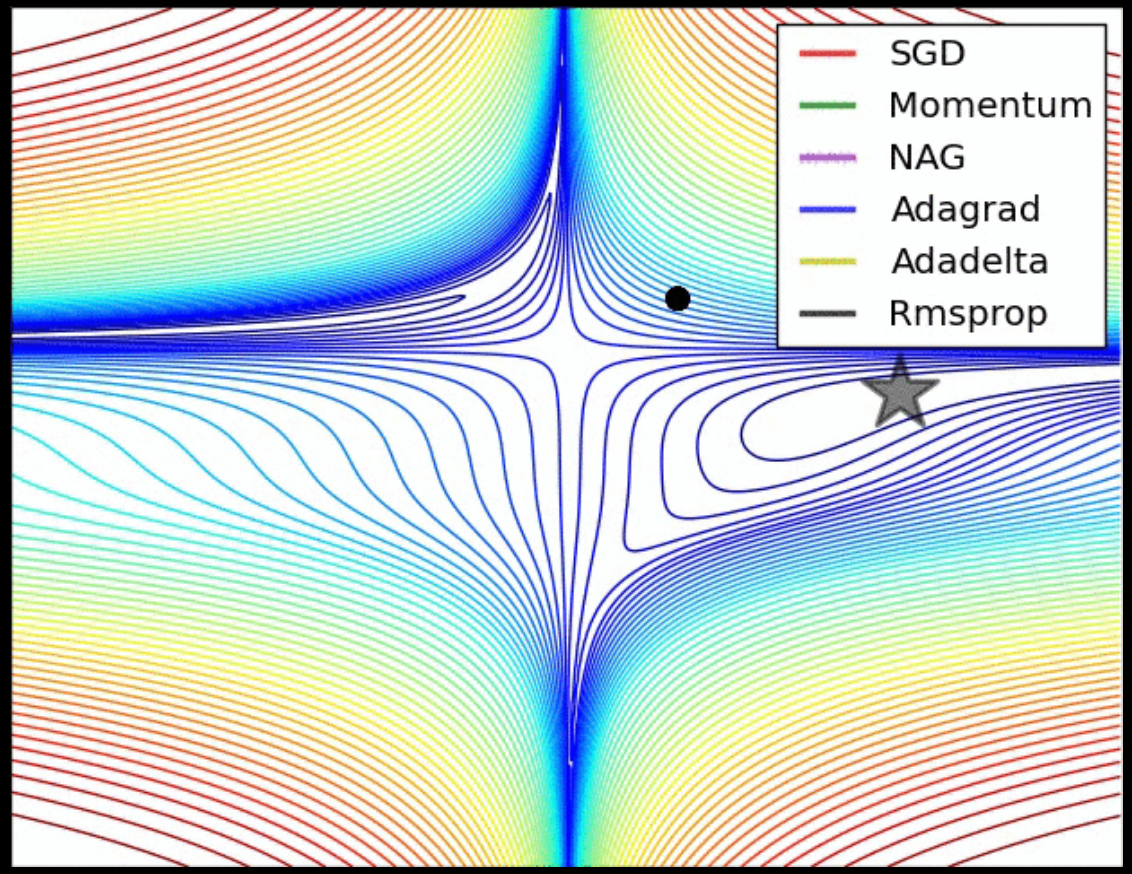
Adam: Adapting Learning Rate Per Parameter

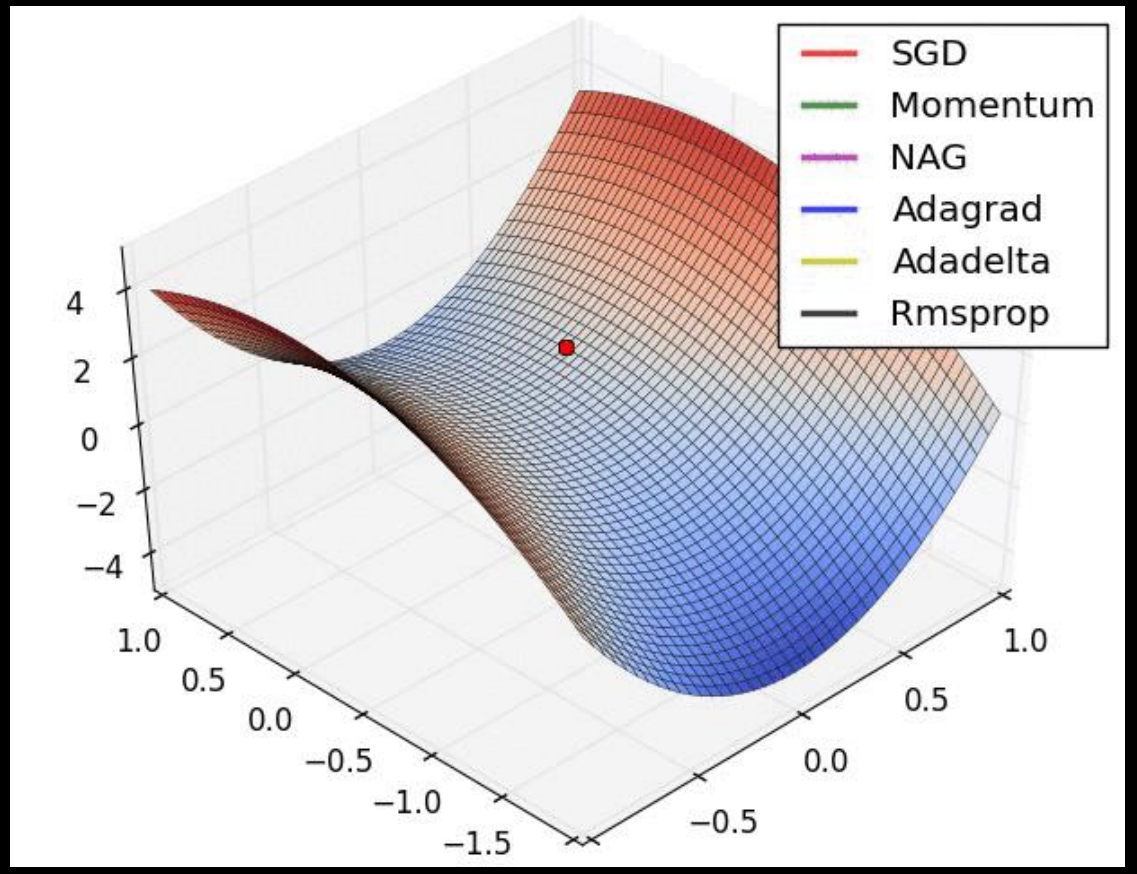
$$W \leftarrow W - \eta \frac{\partial L}{\partial W}$$



Short for “adaptive moment estimation”, Adam dynamically adapts the learning rate for each parameter in a way that appropriately anneals the step size

Adam: Combines AdaGrad and RMSProp





Observation: two methods that adapt the learning rate for each parameter can outperform using a global learning rate for SGD with momentum

http://cs231n.github.io/neural-networks-3/#update; image credit: Alec Radford

Adam: Adapting Learning Rate Per Parameter

Momentum, with hyperparameter controlling decay rate for the gradient's moving average; indicates where to go, with less noisy descent direction from averaging

$$m_t \leftarrow \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$$

Uses moving average of squared gradients to indicate how fast to go? Step size made smaller in steeper directions and larger in flat ones;

$$v_t \leftarrow \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$$

An extra computation overcome issues stemming from initializing these values to 0

Noise

$$W \leftarrow W - \alpha \cdot \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon)$$

Hard-coded value establishing step magnitude's upper bound

"signal-to-noise ratio" (SNR)
- what happens to step size when SNR is smaller?

What optimization method would we get if we had equal values for all parameters' v_t?

Such Learning Tricks Got Neural Networks Deeper,
But Performance Gains Saturated at ~30-40 Layers

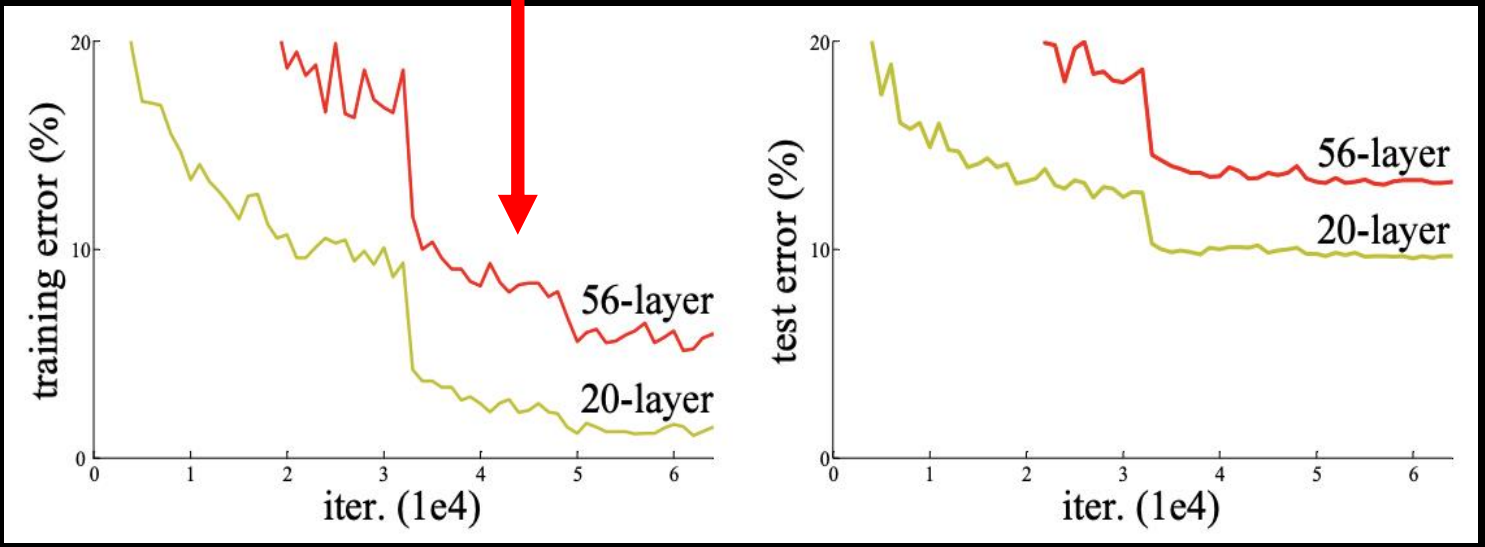
Today's Topics

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- Convergence is hard: He initialization, batch normalization, Adam optimization
- More is needed: residual learning with skip connections (ResNet)
- Programming tutorial

Adding More Layers: What Was the Problem?

- Vanishing gradients?
  - No; with modern tricks at the time (He initialization, BN), gradients well-scaled during training

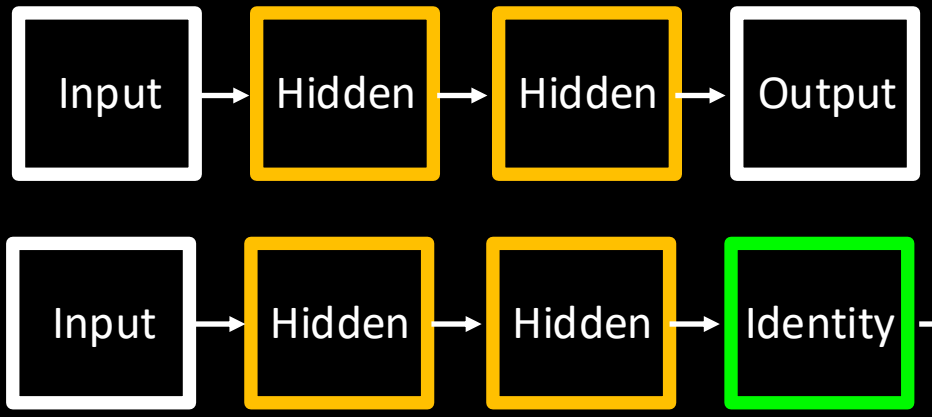
- Overfitting?
  - No; training error is LARGER with more layers

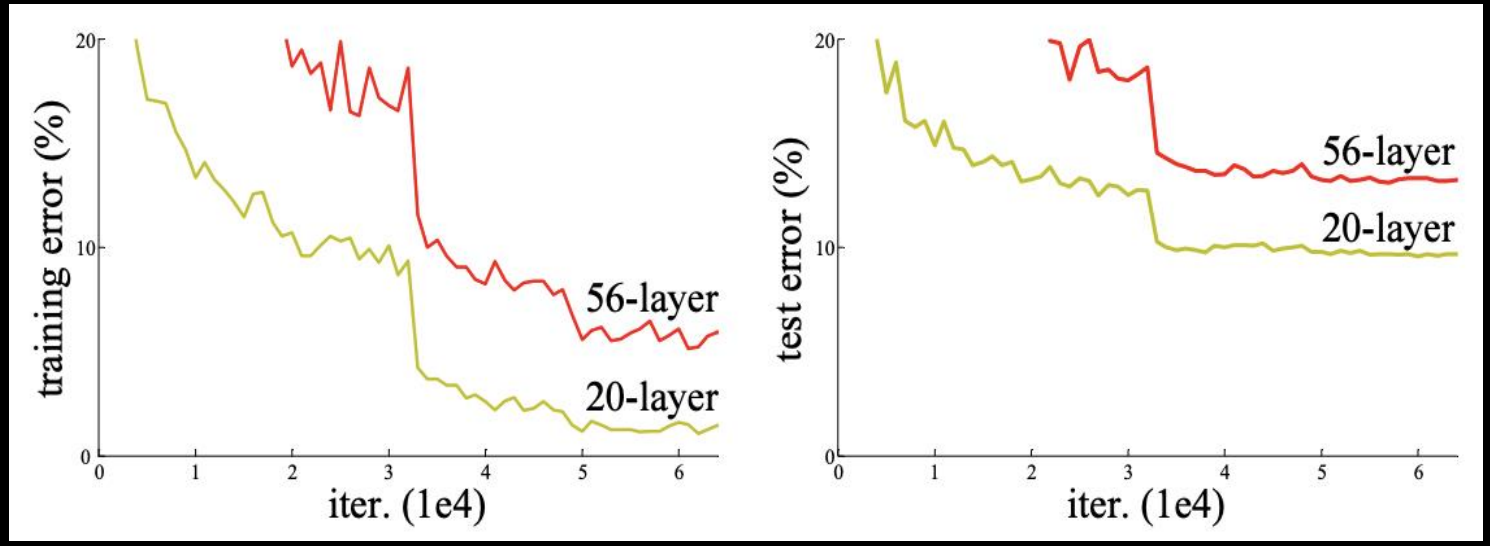


He et al. Deep Residual Learning for Image Recognition. CVPR 2016.

A New Problem! – “Degradation”

- Key question: why do deeper networks perform worse than shallower ones?
- Observation: deeper networks should not be worse than shallower options; extra layers can be “identity” functions

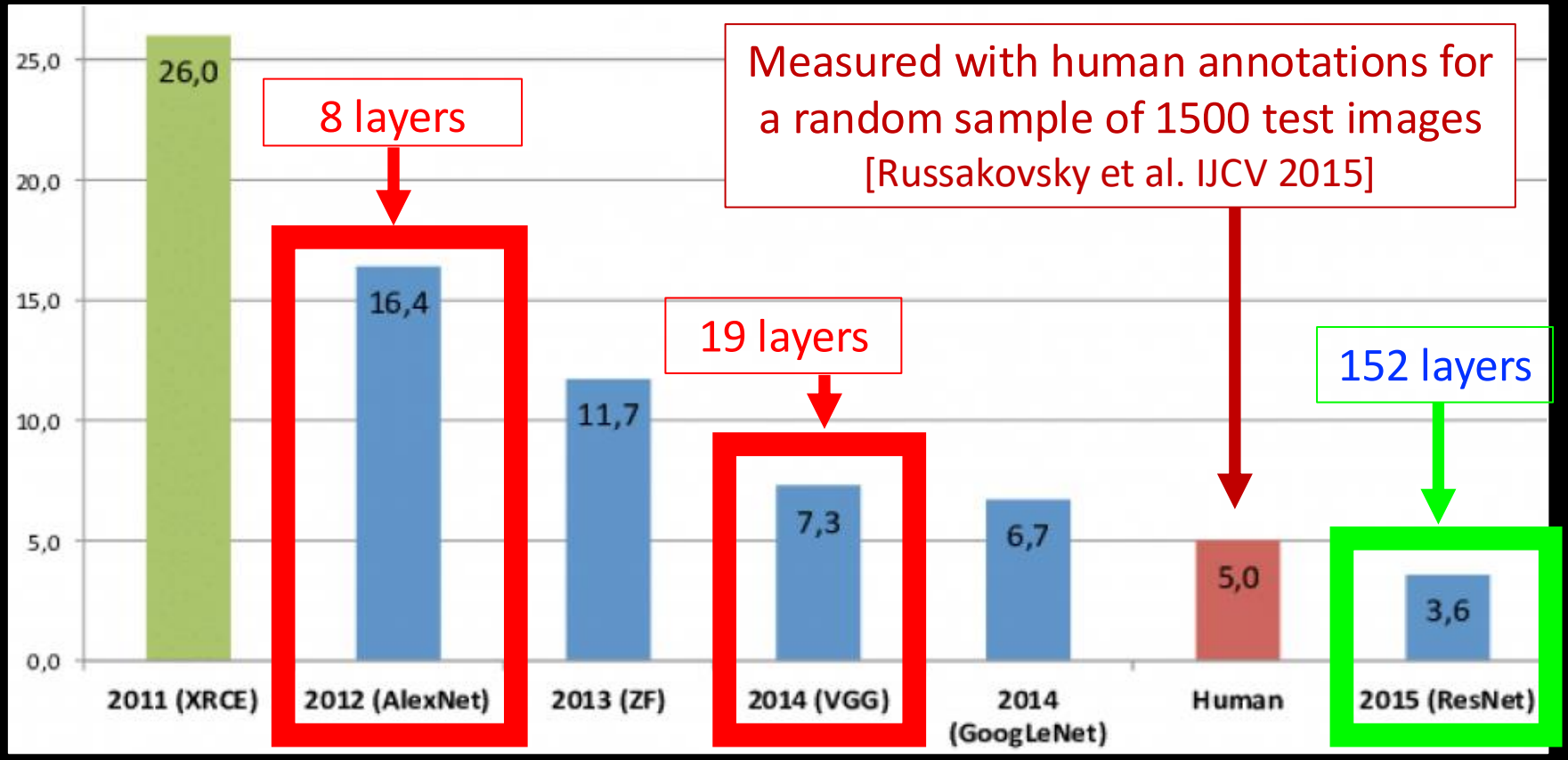




He et al. Deep Residual Learning for Image Recognition. CVPR 2016.

Degradation Problem's Solution: ResNet's Tricks

ImageNet Top 5 Error



For the first time, a model exceeded human-level performance!

https://www.edge-ai-vision.com/2018/07/deep-learning-in-five-and-a-half-minutes/

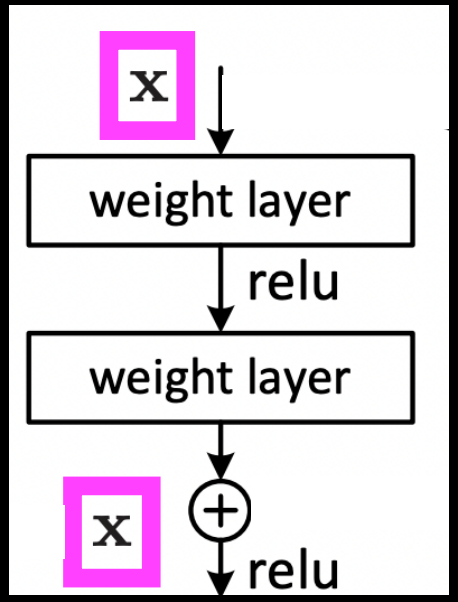
(Model Named After Method)

(2016, CVPR)

Deep Residual Learning for Image Recognition

Kaiming He Xiangyu Zhang Shaoqing Ren Jian Sun
Microsoft Research

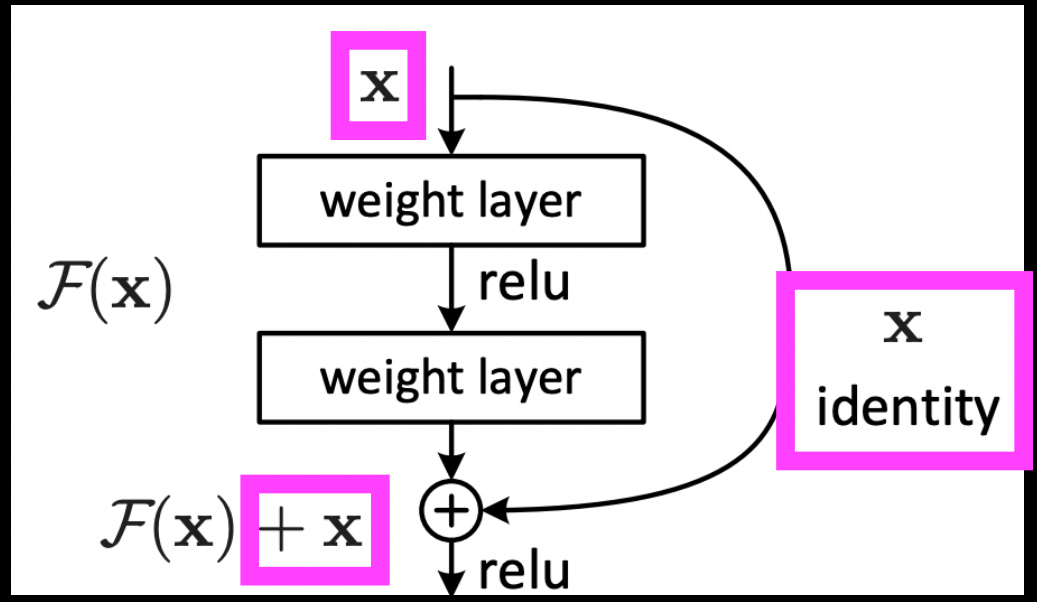
Idea: Facilitate Learning Identity Mapping



Hypothesis: harder to learn a function that performs identity mapping than to learn parameters close to 0

He et al. Deep Residual Learning for Image Recognition. CVPR 2016.

Idea: Facilitate Learning Identity Mapping with Skip Connections



Paper explores when number of layers between skip connections (i.e., residual block) is 2 or 3

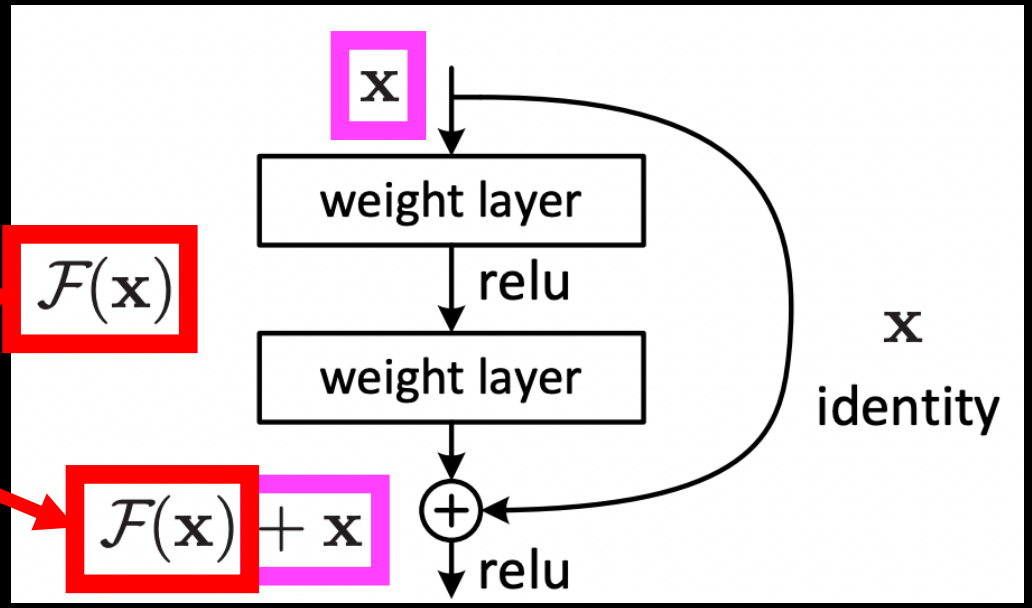
How many model parameters need to be learned when adding a skip connection?

- 0

He et al. Deep Residual Learning for Image Recognition. CVPR 2016.

Idea: Facilitate Learning Identity Mapping with Skip Connections and Residual Learning

"Residual learning" since the function to learn is changed to also include "+x"



Recall: skip connections improve gradient flow since addition has derivative 1, enabling them to flow unchanged

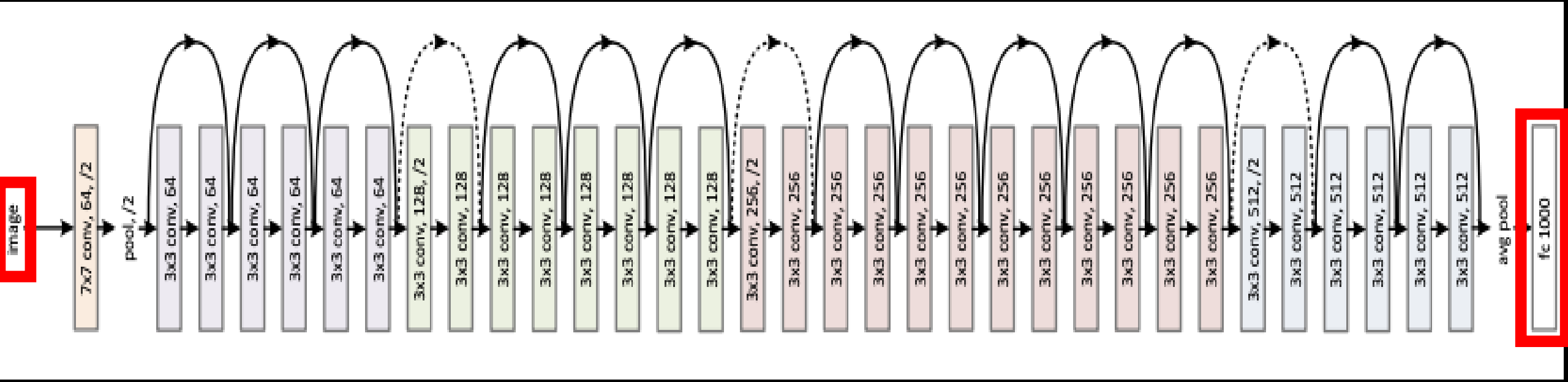
Note: cross-entropy loss optimized for ImageNet classification

What function weights would be learned if the identity mapping is optimal?

e.g., 0; e.g., mean squared error loss function: $\mathcal{L} = (x + F(x) - x)^2$

He et al. Deep Residual Learning for Image Recognition. CVPR 2016.

ResNet Architecture: Shown is Subset of Layers (34 of 152 Layers)



Input: RGB image resized to fixed input size

Output: 1000 class probabilities (sums to 1)

He et al. Deep Residual Learning for Image Recognition. CVPR 2016.

Training Setting: Changes to AlexNet

- Batch size: 256 examples (double amount for AlexNet)
- Initialization: He method
- Momentum optimization: manually adjusted learning rate from initial 0.1 (vs 0.01 for AlexNet), dividing by 10 when validation error plateaued (later experiments used Adam)
- Batch normalization introduced before each convolution's activation
- Weight decay coefficient reduced to 0.0001 (vs 0.005)
- No dropout used in order to isolate analysis on overcoming optimization issues with skip connections and residual learning

Summary: Residual Connections Turned Depth from a Bottleneck to a Design Choice

Today's Topics

- The quest for deeper neural networks: why do they fail?
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- More is needed: residual learning with skip connections (ResNet)
- Programming tutorial

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The End